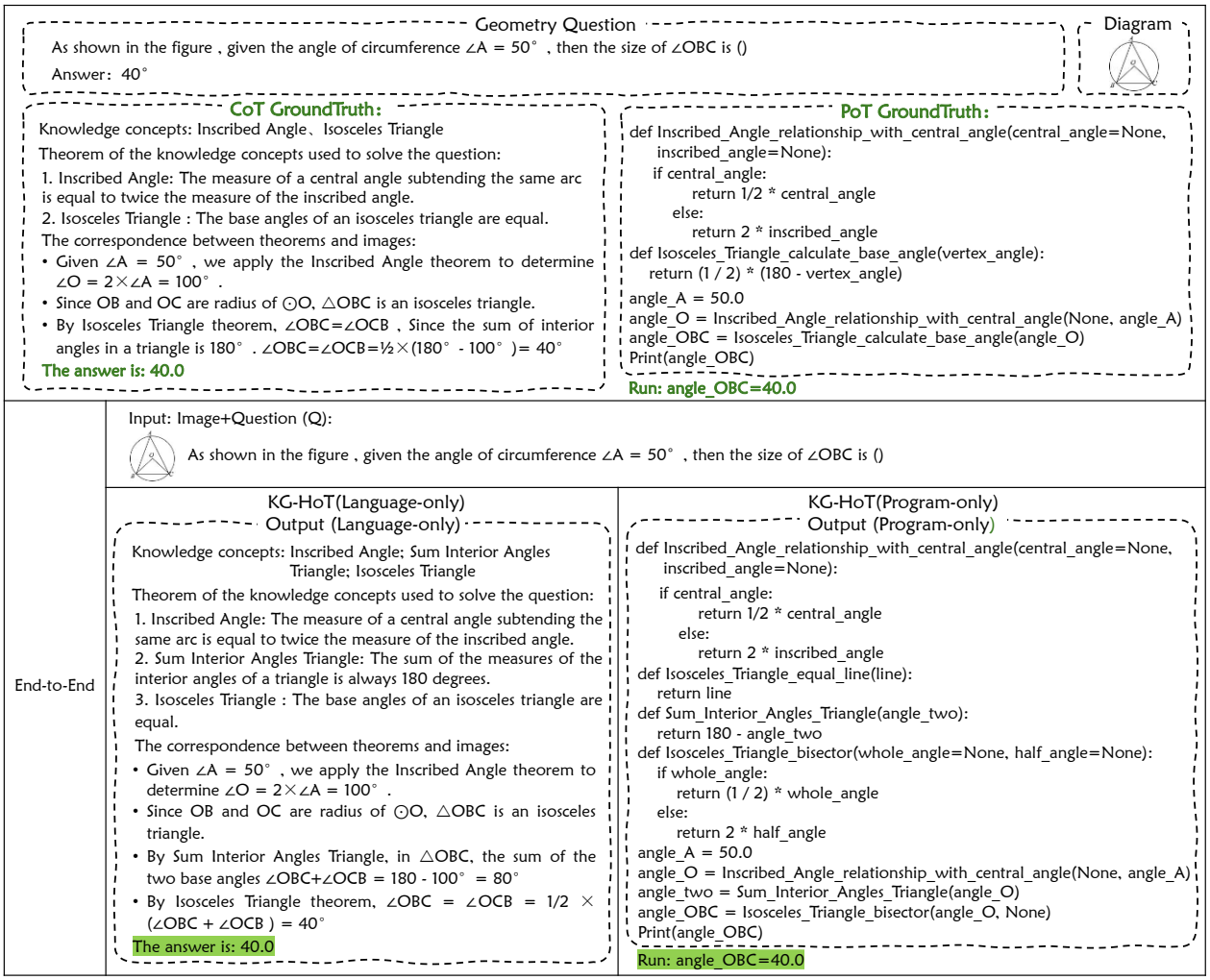

Figure1. Illustration of the end-to-end KG-HoT paradigm. For a given geometry question (top), our model generates both a language-based chain-of-thought (bottom left) and a program-based chain-of-thought (bottom right). Both reasoning chains correctly apply the relevant geometric knowledge to derive the final answer, verifying the effectiveness of our hybrid approach.

<table>
<tr><td rowspan="5">Pipeline</td><td>Stage 1: Q-K</td><td colspan="2">Input: Image+Question (Q):
As shown in the figure , given the angle of circumference ∠A = 50° , then the size of ∠OBC is ()
Output: Knowledge concepts (K): Central Angle; Isosceles Triangle; Sum Interior Angles Triangle
GroundTruth: Inscribed Angle; Isosceles Triangle</td></tr>
<tr><td>Stage 2: QK-PoT</td><td>Input:
Image+Question+Knowledge concepts (QK):
As shown in the figure , given the angle of circumference ∠A = 50° , then the size of ∠OBC is () \n The knowledge concepts involved in this question are: Central Angle; Isosceles Triangle; Sum Interior Angles Triangle</td><td>Output: Program-based chain-of-thought (PoT):
PoT Inference:
def Cenreal_Angle_relationship_with_inscribed_angle(central_angle=None, inscribed_angle=None):
if central_angle:
return 1/2 * central_angle
else:
return 2 * inscribed_angle
def Sum_Interior_Angles_Triangle(angle_two):
return 180 - angle_two
def Isosceles_Triangle_equal_angle(angle):
return angle
angle_A = 50.0
angle_O = Cenreal_Angle_relationship_with_inscribed_angle(angle_A, None)
angle_two = Sum_Interior_Angles_Triangle(angle_O)
angle_OBC = Isosceles_Triangle_equal_angle(angle_two)
Print(angle_OBC)
Run: angle_OBC=155.0</td></tr>
<tr><td>Stage 1: Q-KT</td><td colspan="2">Input: Image+Question (Q):
As shown in the figure , given the angle of circumference ∠A = 50° , then the size of ∠OBC is ()
Output: Knowledge concepts (K): Central Angle; Isosceles Triangle; Inscribed Angle; Equilateral Triangle
Thermos (T): Central Angle: The measure of a central angle subtending the same arc is equal to twice the measure of the inscribed angle; Isosceles Triangle: The two legs of an isosceles triangle are of equal length. The base angles of an isosceles triangle are equal; Inscribed Angle: The measure of a central angle subtending the same arc is equal to twice the measure of the inscribed angle; Equilateral Triangle: All three interior angles are equal to each other and measure 60 degrees.
GroundTruth: Inscribed Angle、Isosceles Triangle
Inscribed Angle: The measure of a central angle subtending the same arc is equal to twice the measure of the inscribed angle; Isosceles Triangle: The base angles of an isosceles triangle are equal;</td></tr>
<tr><td>Stage 2: QKT-PoT</td><td>Input:
Image+Question+Knowledge concepts +Thermos (QKT):
As shown in the figure , given the angle of circumference ∠A = 50° , then the size of ∠OBC is () \n The knowledge concepts involved in this question are: Central Angle; Isosceles Triangle; Inscribed Angle; Equilateral Triangle \n The thermos related to this question are: Central Angle: The measure of a central angle subtending the same arc is equal to twice the measure of the inscribed angle; Isosceles Triangle: The two legs of an isosceles triangle are of equal length. The base angles of an isosceles triangle are equal; Inscribed Angle: The measure of a central angle subtending the same arc is equal to twice the measure of the inscribed angle; Equilateral Triangle: All three interior angles are equal to each other and measure 60 degrees.</td><td>Output: Program-based chain-of-thought (PoT):
PoT Inference:
def Cenreal_Angle_relationship_with_inscribed_angle(central_angle=None, inscribed_angle=None):
if central_angle:
return 1/2 * central_angle
else:
return 2 * inscribed_angle
def Isosceles_Triangle_equal_angle(angle):
return angle
def Inscribed_Angle_relationship_with_central_angle(central_angle=None, inscribed_angle=None):
if central_angle:
return 1/2 * central_angle
else:
return 2 * inscribed_angle
def Equilateral_Triangle_equal_angle():
return 60
angle_A = 50.0
angle_O = Cenreal_Angle_relationship_with_inscribed_angle(angle_A, None)
angle_A = Inscribed_Angle_relationship_with_central_angle(None, angle_O)
Angle_ABC= Isosceles_Triangle_equal_angle(angle_A)
angle_OBC = Equilateral_Triangle_equal_angle()
Print(angle_OBC)
Run: angle_OBC=60.0</td></tr>
<tr><td>Stage 1: Q-CoT</td><td colspan="2">Input: Image+Question (Q):
As shown in the figure , given the angle of circumference ∠A = 50° , then the size of ∠OBC is ()
Output: Language-based chain-of-thought (CoT):
Knowledge concepts: Central Angle; Isosceles Triangle; Inscribed Angle; Sum Interior Angles Triangle; Equilateral Triangle; Theorem of the knowledge concepts used to solve the question: 1. Central Angle: The measure of a central angle subtending the same arc is equal to twice the measure of the inscribed angle; 2. Isosceles Triangle: The two legs of an isosceles triangle are of equal length. The base angles of an isosceles triangle are equal; 3. Inscribed Angle: The measure of a central angle subtending the same arc is equal to twice the measure of the inscribed angle; 4. Sum Interior Angles Triangle: The sum of the measures of the interior angles of a triangle is always 180 degrees; 5. Equilateral Triangle: All three interior angles are equal to each other and measure 60 degrees.
The correspondence between theorems and images: Given ∠A = 50° , we apply the Central Angle theorem to determine ∠O = 1/2 ×∠A = 25° ; Since AB=AC in the figure, △ABC is an isosceles triangle; By Isosceles Triangle theorem, ∠ABC=∠ACB; we apply the Inscribed Angle theorem to determine ∠BAC = 2 ×∠O = 50° ; By Sum Interior Angles Triangle, in △ABC, the sum of the two base angles ∠ABC+∠ACB = 180 - 50° = 130° ; By Isosceles Triangle theorem, ∠ABC=∠ACB= 1/2 ×∠BAC = 65° ; In △OBC, OB=OC=BC, thus △OBC is Equilateral Triangle, thus ∠OBC=60°
The answer is: 60.0
GroundTruth: CoT GroundTruth</td></tr>
<tr><td>Stage 2: QCoT-PoT</td><td>Input:
Image+Question+ Language-based chain-of-thought (QCoT):
As shown in the figure , given the angle of circumference ∠A = 50° , then the size of ∠OBC is () \n The Language-based chain-of-thought involved in this question is: Knowledge concepts: Central Angle; Isosceles Triangle; Inscribed Angle; Sum Interior Angles Triangle; Equilateral Triangle; Theorem of the knowledge concepts used to solve the question: 1. Central Angle: The measure of a central angle subtending the same arc is equal to twice the measure of the inscribed angle; …… 4. Sum Interior Angles Triangle: The sum of the measures of the interior angles of a triangle is always 180 degrees; 5. Equilateral Triangle: All three interior angles are equal to each other and measure 60 degrees. The correspondence between theorems and images: …… we apply the Inscribed Angle theorem to determine ∠BAC = 2 ×∠O = 50° ; By Sum Interior Angles Triangle, in △ABC, the sum of the two base angles ∠ABC+∠ACB = 180 - 50° = 130° ; By Isosceles Triangle theorem, ∠ABC=∠ACB= 1/2 ×∠BAC = 65° ; In △OBC, OB=OC=BC, thus △OBC is Equilateral Triangle, thus ∠OBC=60°
The answer is: 60.0</td><td>Output: Program-based chain-of-thought (PoT):
PoT Inference:
def Cenreal_Angle_relationship_with_inscribed_angle(central_angle=None, inscribed_angle=None):
if central_angle:
return 1/2 * central_angle
else:
return 2 * inscribed_angle
def Isosceles_Triangle_equal_angle(angle):
return angle
def Isosceles_Triangle_bisector(whole_angle=None, half_angle=None):
if whole_angle:
return (1 / 2) * whole_angle
else:
return 2 * half_angle
def Inscribed_Angle_relationship_with_central_angle(central_angle=None, inscribed_angle=None):
if central_angle:
return 1/2 * central_angle
else:
return 2 * inscribed_angle
def Sum_Interior_Angles_Triangle(angle_two):
return 180 - angle_two
def Equilateral_Triangle_equal_angle():
return 60
angle_A = 50.0
angle_O = Cenreal_Angle_relationship_with_inscribed_angle(angle_A, None)
angle_BAC = Inscribed_Angle_relationship_with_central_angle(None, angle_O)
angle_two = Sum_Interior_Angles_Triangle(angle_BAC)
angle_ABC = Isosceles_Triangle_bisector(angle_two, None)
angle_ACB= Isosceles_Triangle_equal_angle(angle_ABC)
angle_OBC = Equilateral_Triangle_equal_angle()
Print(angle_OBC)
Run: angle_OBC=60.0</td></tr>
</table>

Figure 2. Pipeline paradigms targeting program-based chain-of-thought generation. The figure illustrates three variants of the two-stage pipeline approach. In each variant, the purple-highlighted content is the intermediate output from the first stage, which is concatenated with the original input to form the input for the second stage. As shown, all three pipeline variants fail to derive the correct answer for this geometry problem, highlighting the limitations of the decomposed pipeline approach compared to the proposed end-to-end paradigm.

Geometry Question

As shown in the figure , given the angle of circumference ∠A = 50° , then the size of ∠OBC is ()

Answer: 40°

Diagram

CoT GroundTruth:

Knowledge concepts: Inscribed Angle、Isosceles Triangle

Theorem of the knowledge concepts used to solve the question:

1. Inscribed Angle: The measure of a central angle subtending the same arc is equal to twice the measure of the inscribed angle.
2. Isosceles Triangle : The base angles of an isosceles triangle are equal.

The correspondence between theorems and images:

- Given ∠A = 50° , we apply the Inscribed Angle theorem to determine ∠O = 2×∠A = 100° .
- Since OB and OC are radius of ⊙O, △OBC is an isosceles triangle.
- By Isosceles Triangle theorem, ∠OBC=∠OCB , Since the sum of interior angles in a triangle is 180° . ∠OBC=∠OCB=½×(180° - 100°)= 40°

The answer is: 40.0

PoT GroundTruth:

```
def Inscribed_Angle_relationship_with_central_angle(central_angle=None,
    inscribed_angle=None):
    if central_angle:
        return 1/2 * central_angle
    else:
        return 2 * inscribed_angle
def Isosceles_Triangle_calculate_base_angle(vertex_angle):
    return (1 / 2) * (180 - vertex_angle)
angle_A = 50.0
angle_O = Inscribed_Angle_relationship_with_central_angle(None, angle_A)
angle_OBC = Isosceles_Triangle_calculate_base_angle(angle_O)
Print(angle_OBC)
```

Run: angle_OBC=40.0

Pipeline

Stage 1: Q-PoT

Input:

Image+Question (Q):

As shown in the figure , given the angle of circumference ∠A = 50° , then the size of ∠OBC is ()

Output: Program-based chain-of-thought (PoT):

PoT Inference:

```
def Isosceles_Triangle_equal_line(line):
    return line
def Isosceles_Triangle_equal_angle(angle):
    return angle
angle_A = 50.0
angle_BC = 180 – angle_A
angle_B = (1 / 2) * angle_BC
angle_C = Isosceles_Triangle_equal_angle(angle_B)
angle_O = 90.0
angle_BC = 180 - angle_O
angle_OBC = (1 / 2) * angle_BC
Print(angle_OBC)
```

Run: angle_OBC=45.0

GroundTruth: PoT GroundTruth

Stage 2: QPoT-CoT

Input:

Image+Question+Program-based chain-of-thought (QPoT):

As shown in the figure , given the angle of circumference ∠A = 50° , then the size of ∠OBC is () \n

```
def Isosceles_Triangle_equal_line(line):
    return line
def Isosceles_Triangle_equal_angle(angle):
    return angle
angle_A = 50.0
angle_BC = 180 – angle_A
angle_B = (1 / 2) * angle_BC
angle_C = Isosceles_Triangle_equal_angle(angle_B)
angle_O = 90.0
angle_BC = 180 - angle_O
angle_OBC = (1 / 2) * angle_BC
Print(angle_OBC)
```

Output: Language-based chain-of-thought (CoT):

CoT Inference:

Knowledge concepts: Isosceles Triangle

Theorem of the knowledge concepts used to solve the question: 1. Isosceles Triangle: The two legs of an isosceles triangle are of equal length. The base angles of an isosceles triangle are equal.

The correspondence between theorems and images: Given ∠A = 50° , In △ABC, AB=AC, thus △ABC is Isosceles Triangle, ∠B+∠C = 180° - ∠A = 180 - 50° = 130° ; ∠B = ∠C = 1/2×(∠B+∠C) = 1/2 × 130° =65° ; In △OBC, ∠O = 90° , and OB = OC, thus △OBC is Isosceles Triangle, ∠B+∠C = 180° - ∠O = 180 - 90° = 90° ; ∠B = ∠C = 1/2×(∠B+∠C) = 1/2 × 90° = 45° ; ∠OBC = ∠B = 45°

The answer is: 45.0

Figure 3. Pipeline paradigm targeting language-based chain-of-thought generation. The two-stage pipeline first generates a program-based chain-of-thought from the original image and question (Stage 1), then uses this intermediate output to derive the final language-based reasoning chain (Stage 2). As shown, this pipeline approach fails to produce the correct answer, highlighting the limitations of decomposed two-stage reasoning compared to the proposed end-to-end KG-HoT paradigm.